

Jonathan J. Froehlich

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Experimental-computational scientist who builds scalable perturbation and functional-genomics approaches to answer complex biological questions.

PROFESSIONAL EXPERIENCE

[\[LinkedIn\]](#)

2023–2026

Postdoctoral researcher

Max Planck Institute for Molecular Genetics, Berlin, DE, Systems Epigenetics, Dr. Edda Schulz.

- Integrated quantitative Oct4 CRISPRi perturbations with scRNA-seq and ATAC-seq to identify dose-responsive gene programs and candidate regulatory DNA in mouse embryonic stem cells.
- Established experimental and computational workflows spanning scRNA-seq, ATAC-seq, and MPRA to prioritize and test cis-regulatory elements across transcription-factor doses.

2014–2021

Doctoral researcher

Max Delbrueck Center for Molecular Medicine & Berlin Inst. for Medical Systems Biol., Berlin, DE, Systems Biology of Regulatory Elements, Prof. Nikolaus Rajewsky

- Developed an *in vivo*, parallelized CRISPR-Cas9 screening in *C. elegans* to test >1,000 gene regulatory mutations, a 100-fold increase over previous approaches.
- Designed a kilobase-scale amplicon-sequencing protocol and partnered with a computational collaborator to build specialized analysis software for high-throughput regulatory screens.
- Discovered that adjacent microRNA binding sites in the *lin-41* 3'-UTR show independent and compensatory regulatory effects.

2012, 2010

Bachelor's & Master's student

Max Planck Institute of Immunobiology and Epigenetics, Freiburg, DE, Epigenetics, Prof. Thomas Jenuwein

- Developed a semi-automated microscopy and image-analysis workflow for quantifying heterochromatin state in reporter cells.
- Established a microscopy-based screen with a kinase inhibitor library to demonstrate that the PI3K signaling pathway promotes cellular heterochromatin condensation.

EDUCATION

2022 **Dr. rer. nat., Molecular Biology**, Humboldt University, Berlin, DE

2012 **MSc, Biology**, Ludwig Maximilian University, Munich, DE

2010 **BSc, Molecular Biotechnology**, FH Campus Vienna, Vienna, AT

TECHNICAL EXPERIENCE

LABORATORY

2023–2026	Tunable CRISPRi, ATAC-seq, scRNA-seq, MPRA
2023–2026	Cell culture of embryonic stem cells
2015–2026	Synthetic DNA construct design, cloning, delivery, line generation
2009–2026	Epifluorescence and confocal microscopy; flow cytometry
2007–2026	Molecular biology of RNA, DNA, and proteins
2014–2022	Multiplexed CRISPR-Cas9 screening of noncoding DNA in <i>C. elegans</i>
2014–2022	<i>C. elegans</i> handling and phenotype screening
2017–2021	mRNA/DNA amplicon sequencing Illumina & PacBio
2010, 2012	96-well semi-automated microscopy-based compound screening
2011–2012	<i>D. melanogaster</i> handling and maintenance
2009–2011	Cell culture of fibroblast cell lines and embryonic stem cells

COMPUTATIONAL

2024–2026	ATAC-seq and motif-grammar analysis; ChromBPNet; TF motifs
2023–2026	scRNA-seq analysis with STARsolo, R, and Bioconductor
2022–2026	Reproducible analysis of RT-qPCR, FACS, and other data with R
2014–2026	Collaborated in interdisciplinary experimental-computational teams
2007–2026	Adobe Illustrator/Photoshop, Geneious/ApE/BioEdit
2010, 2012	Automated image analysis with CellProfiler, and CPAnalyst/SQL
2008, 2020	3D imaging: Amira for segmentation, Blender to measure volumes

PUBLICATIONS

[Google Scholar] [ORCID]

- 2025 Till Schwämmle, Gemma Noviello, Eleni Kanata, **Jonathan J Froehlich**, Melissa Bothe, Aybuge Altay, Jade Scouarnec, Vivi-Yun Feng, Martin Vingron, Edda G Schulz, “Reporter CRISPR screens decipher cis- and trans-regulatory principles at the Xist locus”, *Nature Structural & Molecular Biology*, [DOI].
- 2023 **Jonathan J Froehlich**, Nikolaus Rajewsky, “*C. elegans* gene regulatory alleles and reporter bashing studies”, *Micropublication Biology*, [DOI].
- 2022 **Jonathan J Froehlich**, “Parallel Genetics of Gene Regulatory Sequences in *Caenorhabditis elegans*”, Dissertation, Humboldt University Berlin, [DOI].
- 2021 **Jonathan J Froehlich***, Bora Uyar*, Margareta Herzog, Kathrin Theil, Petar Glažar, Altuna Akalin, Nikolaus Rajewsky, “Parallel genetics of regulatory sequences using scalable genome editing in vivo”, *Cell Reports*, [DOI] [news] [software] *co-first
- 2021 **Jonathan J Froehlich**, Nikolaus Rajewsky, Collin Ewald, “Estimation of *C. elegans* cell- and tissue volumes”, *Micropublication Biology*, [DOI].
- 2013 Sylvain Maenner, Marisa Müller, **Jonathan Froehlich**, Diana Langer, Peter B Becker, “ATP-dependent roX RNA remodeling by the helicase maleless enables specific association of MSL proteins”, *Molecular Cell*, [DOI].
- 2012 Silvia Dambacher, Wen Deng, Matthias Hahn, Dennis Sadic, **Jonathan Froehlich**, Alexander Nuber, Christian Hoischen, Stephan Diekmann, Heinrich Leonhardt, Gunnar Schotta, “CENP-C facilitates the recruitment of M18BP1 to centromeric chromatin”, *Nucleus*, [DOI].

ACTIVITIES

LEADERSHIP & ENGAGEMENT

2023–2026	Postdoc association representative
2023–2026	Maintainer of team's website
2023–2026	Led one main- and collaborated on five side projects
2015, 2023, 2025	Co-organized three research retreats for 15-30 participants
2023	Co-hosted a distinguished speaker at the institute
2014–2023	Contributed [17 biological diagrams] to Wikipedia
2023	Designed the program booklet and supported conference organization
2014–2022	Led one main- and two side projects from conceptualization to publication
2020–2021	Coordinated weekly team meetings, established guidelines
2020–2021	Initiated a community gardening project on the institute's terrace
2016–2021	Coordinated monthly social event "After Hour" at the institute, MDC/BIMSB
2014–2019	Communicated science to public at 4 events "Long Night of Science Berlin"

SIDE PROJECTS

2026	[Papers-to-table] extracts information from scientific publications
2026	[Curated resources] for research work, communication, lab culture
2025	[A game] that explores limitations of citation counts

TEACHING

04/2024–09/2025	Supervised Master's student, CRISPRi lines and cis-regulatory DNA reporter
06/2018–07/2018	Supervised Master's student, knock-in <i>C. elegans</i>
02/2023	Instructor at university course on CRISPR, Free University, Berlin
03/2014	Instructor at university course on <i>C. elegans</i> , Humboldt University, Berlin

SELECTED PRESENTATIONS

2025	Institute Seminar, Max Planck Institute for Molecular Genetics, Berlin
2019	Selected Talk, 22nd International <i>C. elegans</i> Conference, Los Angeles
2017	Selected Talk, 21st International <i>C. elegans</i> Conference, Los Angeles
2016	Poster, EMBO "Complex Life of mRNA", Heidelberg
2015/2017/2019	Talks, Student Seminar Series, Max Delbrueck Center, Berlin